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TECHNOLOGY**

NED SMART EXAMINATION MANAGEMENT SYSTEM (NED-SEMS)

BY

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Abstract

At present, the examination process within the Northern Education Division (NED) relies on a manual and semi-digital workflow that begins at the Education Division Manager (EDM) office which appoints subject teachers from various schools to serve as item writers [1]. These teachers prepare draft examination papers based on the national syllabus and submit them to the EDM office for moderation, where other teachers review the content for accuracy and fairness. Approved papers are then distributed to Headteachers via email or WhatsApp, who pass them to school examination officers for printing and distribution. After students complete the exams, teachers manually mark scripts, compile results, and forward them up the chain through the Headteacher to the EDM office for consolidation and official release. This process is time-consuming, error-prone, and lacks centralized control, transparency, and analytical insight.

To address these challenges, the NED Smart Examination Management System (NED-SEMS) is proposed as an intelligent web-based platform that will automate and secure examination workflows by supporting role-based access for all key actors ensuring secure exam paper creation, moderation, controlled distribution, and results submission. It will provide real-time dashboards, automated report generation, and an AI module capable of detecting irregularities and predicting performance trends which will be developed using PHP and MySQL for core system operations, Python for AI analytics, and standard HTML, CSS, and JavaScript for the user interface. NED-SEMS enhances efficiency, transparency, and accountability in the management of examinations by using Agile development methodology, the system will evolve through iterative refinement and continuous user feedback, transforming the Division's fragmented processes into an intelligent, centralized, and data-driven examination management environment.

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1.0 Introduction

NED operates under the Ministry of Education, Science and Technology (MoEST) which oversees the coordination, supervision, and standardization of education and examinations across secondary and primary schools in the Northern Region of Malawi [1]. To strengthen transparency, accountability, and efficiency in these processes, the NED Smart Examination Management System (NED-SEMS) will be developed as a web-based platform that automates and secures the management of examination activities across the division. It will provide a centralized platform that will support all key stages of examination management including question paper preparation, moderation, distribution, marking and official release of results. Through its modular design, NED-SEMS will ensure that examination workflows are transparent, traceable, and efficiently executed across all schools within the division.

The system will adopt a role-based access model that gives distinct privileges to system actors, where each user interacts with the system through a secure login interface and perform actions consistent with their responsibilities. NED-SEMS will include automated notifications, document version tracking, and analytical dashboards that will enable education administrators to monitor progress, identify anomalies and generate real-time performance reports. It will also apply modern software design principles that emphasize usability, scalability, and data security, with the goal of improving examination integrity by reducing manual handling of sensitive materials, enhancing accountability, and minimizing risks of data loss or leakage. Despite these capabilities, the Division's current examination process still faces operational challenges that hinder efficiency and data management, which the proposed system is designed to address.

2.0 Problem statement

The current examination workflow in the Northern Education Division (NED) relies heavily on manual document handling and informal digital tools such as email, WhatsApp and spreadsheets which creates a fragmented and uncoordinated process in which draft examinations, moderation comments and students' results are shared across multiple disconnected channels [2]. Research by Korte shows that such fragmented information flows lead to inconsistent processes, frequent errors, and weak traceability and Guidone further explains that manual document handling increases the likelihood of data loss, delayed communication, and exposure of sensitive information which becomes a significant problem because examinations require accuracy, confidentiality and strict accountability[10][11]. Without a coordinated and secure system, the risk of unauthorized access, accidental changes, and delayed feedback remains high hence threaten the fairness, reliability, and overall integrity of the examination process in the division.

3.0 Scope of the Development Project

The NED Smart Examination Management System (NED-SEMS) defines the boundaries and functionalities of a centralized digital platform that will manage all examination-related activities across the Northern Education Division (NED). The system will focus much on automating exam preparation, moderation, distribution, performance tracking, and decision-making processes while maintaining strict data security and accountability standards.

3.0.1 Included features

3.0.1.1 User Management

The User Management module will control access, authentication, and accountability of users within the NED-SEMS by assigning role-based privileges to Education Division Managers, Headteachers, Examination Officers and Teachers so that each user executes only actions appropriate for their assigned role. The module will again provide secure registration and login allowing users to update their profiles and records all activity via session tracking and system-generated logs. These logs form a complete audit trail that enhances transparency, supports investigations, and improves overall accountability.

3.0.1.2 Examination Management

This Examination Management module will allow safe storage of all examination's documents within NED-SEMS by letting item writers and exam officers upload exam papers to the system, then moderators and division officers can check and approve them in an organized manner. The module will have version control to show any updates and changes for audit. It will also create exam timetables automatically based on subject information and school schedules and when approved, exam papers will be shared through secure methods to keep them private. The system will also use AI to detect similar or copied content, which will help prevent exam leaks and make sure exam materials are secure and reliable.

3.0.1.3 Performance and Reporting management

The Performance and Reporting Management module will handle the collection, organization, and analysis of examination data across all schools in the Northern Education Division. It will gather results submitted through the system, consolidate them into structured datasets, and present them through clear visual summaries such as charts, trend indicators, and comparative analyses. The module will enable the generation of standardized and customized reports based on school, subject, district, or time period. It will also support real-time monitoring by showing the progress of script submissions, marking activities, and compliance with examination timelines. By providing structured performance visibility and traceable reporting processes, this module will help education managers understand the state of examination outcomes before making further decisions.

3.0.1.4 Prediction and decision support

The Prediction and Decision Support module will go beyond simple reports, It will use AI to find patterns, warn of danger, and help with planning by using machine learning to check problems like sudden changes in scores and potential test leaks. It will also make predictions on how scores might change based on past data and tell which schools might need more help or attention. It will also give tips to improve decisions, use resources better, and plan actions and rules in the division. With these prediction and analysis tools, the module will help leaders make smarter, more ready, and fact-based choices.

3.1 Excluded Features

- Online student examinations (live assessments).
- Integration with external learning platforms.
- Mobile application development.

3.2 Before Installation

Before NED-SEMS, exam management was scattered where exam papers were created by item writers and saved on local computers or printed for review, while moderation was done separately with input stored in another document shared via email or messaging apps. Exam schedules and timetables were put together by each school and sent to staff via printed notices or messages. Results were collected in spreadsheets made at the school level and then combined manually at the divisional level, which was slow and sometimes incorrect. Communication among teachers, exam staff, and officials was done through calls, emails, and messaging apps, without a clear way to monitor who was doing what or when it was finished. Also, different software and personal storage devices used by users created problems with compatibility and could lead to lost or hacked files. This creates the starting situation for the new system, showing how the work is done, what tools are used, and who does what to keep the system balanced and working, without restating the issues talked about in section 2.

3.3 After installation

Once implemented, NED-SEMS will digitize and centralize all examination activities with role-based access. Exam papers will be uploaded, moderated, and approved in a controlled digital environment, ensuring transparency and traceability. Sensitive information will stay secure, and results will be entered and compiled electronically for real-time analytics [8]. AI-driven tools will also monitor irregularities, predict performance trends, and generate actionable insights. Automated alerts will facilitate timely coordination, better decision-making, and enhanced accountability across all schools in the division [4].

3.4 System Integration Considerations

At present, digital support for these tasks is limited, primarily involving spreadsheets and fragmented record-keeping. NED-SEMS aims to offer a comprehensive, self-sufficient framework capable of replacing manual

and semi-digital methods. It will enable the integration of past records to prevent loss and simplify adoption from current systems. NED-SEMS will facilitate unified oversight of examination processes while preserving data accuracy, confidentiality, and integration with common office applications [9].

4.0 System Personnel Future

The successful operation and sustainability of the NED Smart Examination Management System depend on a clearly defined structure of personnel responsible for its use, management, and continued enhancement. The system involves three main categories of personnel: end users, project team, and owners.

4.1 System End Users

These are the main individuals who interact with the NED Smart Examination Management System in daily and periodic operations.

- **Education Division Manager (EDM):**
Oversees all examination activities, appoints teachers as item writers or moderators, and monitors overall progress through administrative dashboards and reports.
- **Appointed Teachers:**
Selected subject teachers who prepare examination papers, upload drafts with marking schemes, and later review and approve papers during moderation to ensure accuracy and fairness.
- **HeadTeachers:**
Supervise examination processes at school level, access approved papers, coordinate printing and administration, and oversee result submission and reporting.
- **Examination Officers:**
Handle school-level logistics such as printing, distributing, and compiling examination materials, then submit verified data electronically to the division.
- **ICT Officer:**
Provide technical support, manage user access, and ensure system reliability across schools and the division.

4.2 Project Team

Student (Leonard Ponje Mlungu): Responsible for the end-to-end execution of the NED Smart Examination Management System project, including requirement analysis, system design, front-end and back-end development, database architecture and administration, API integration, and implementation of AI-driven recommendation logic. Additionally, oversees quality assurance, manages project timelines, coordinates deliverables, and ensures alignment with the project scope in consultation with the supervisor.

Supervisor (Mr. John Memory Banda): Provides strategic guidance and academic oversight throughout the project, reviewing progress, ensuring methodological rigor, facilitating access to resources, and validating the system against its defined objectives and requirements.

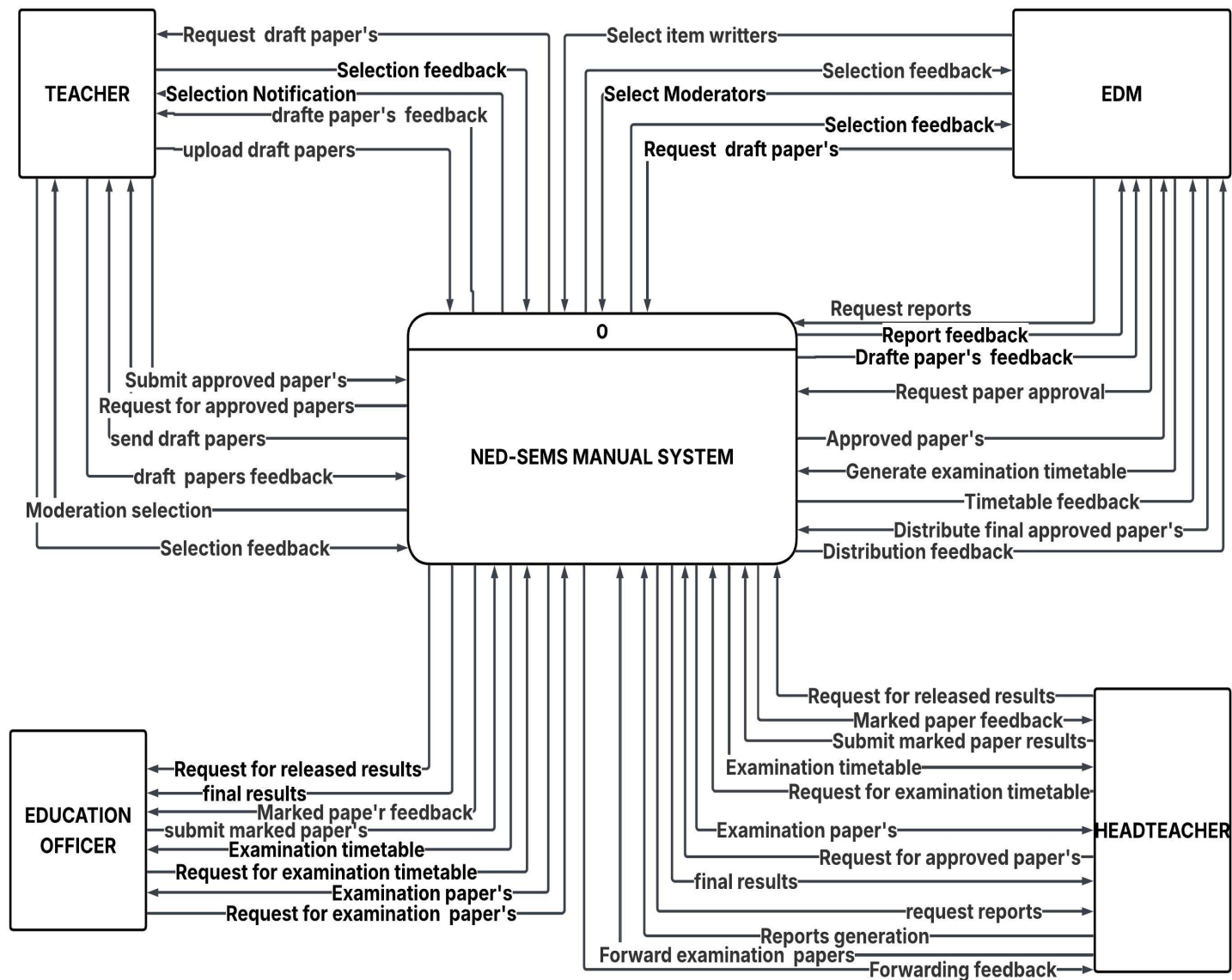
4.3 System Owner

The system owner for the NED Smart Examination Management System is the student developer, Leonard Ponje Mlungu, under the academic supervision of Mr. John Memory Banda from the Department of Education in ICT at the University of Livingstonia. As system owner, the student is responsible for developing and managing the system within the scope of the project, while the supervisor provides guidance, ensures adherence to academic standards, and validates that the system meets the defined requirements.

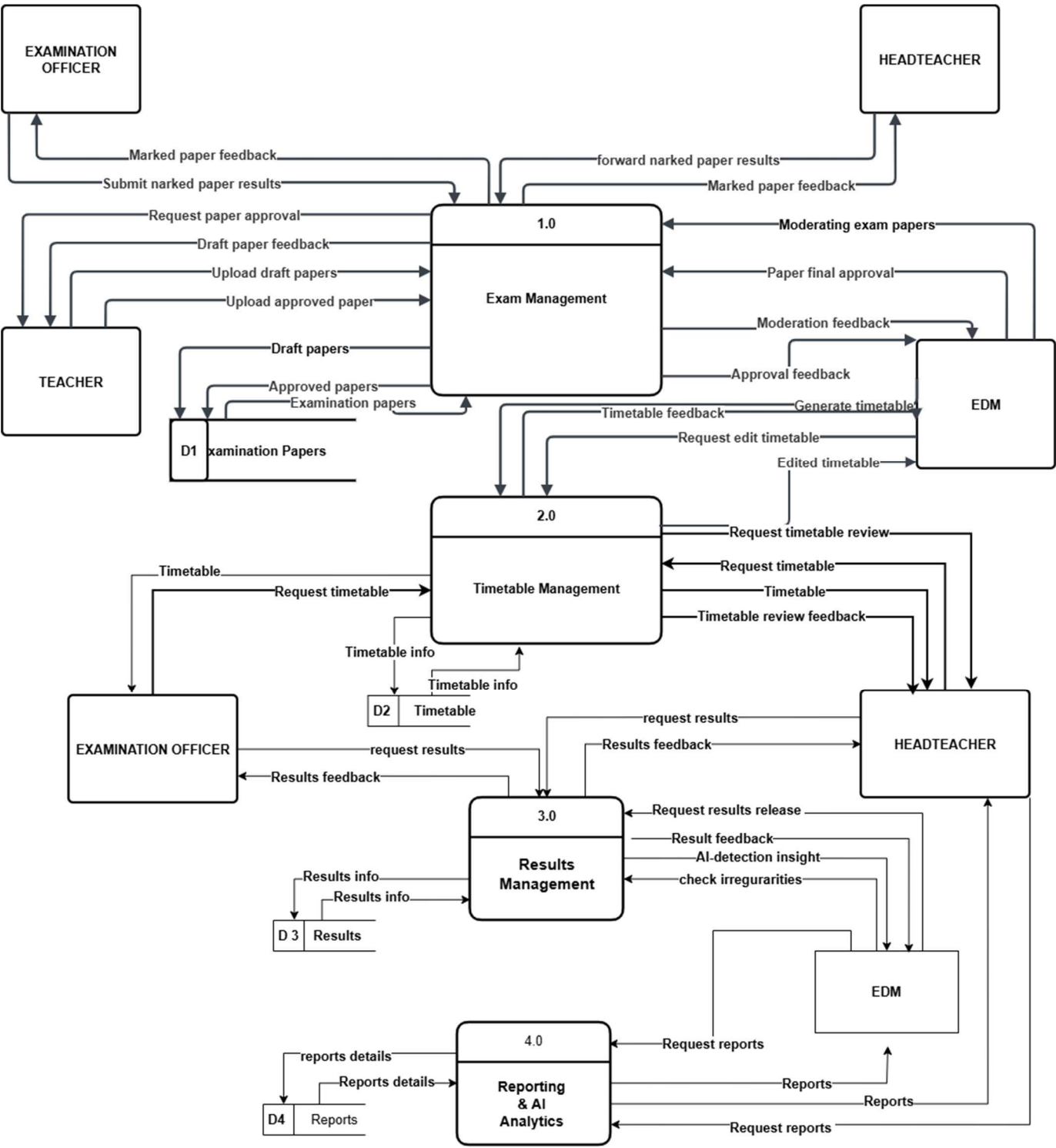
5.0 System Design Overview

5.1 DFD

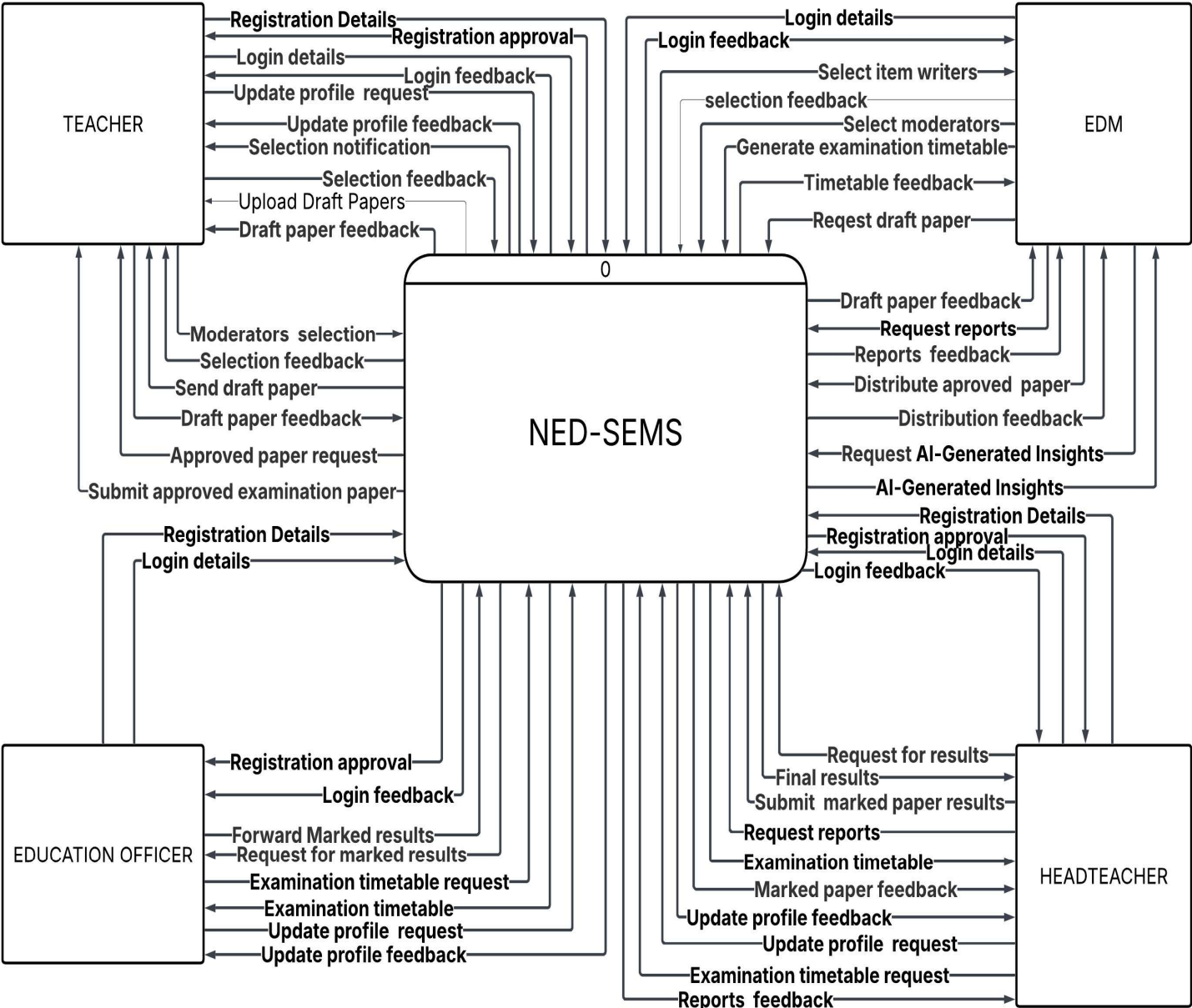
5.1.1 Figure 1: Context diagram for the current system



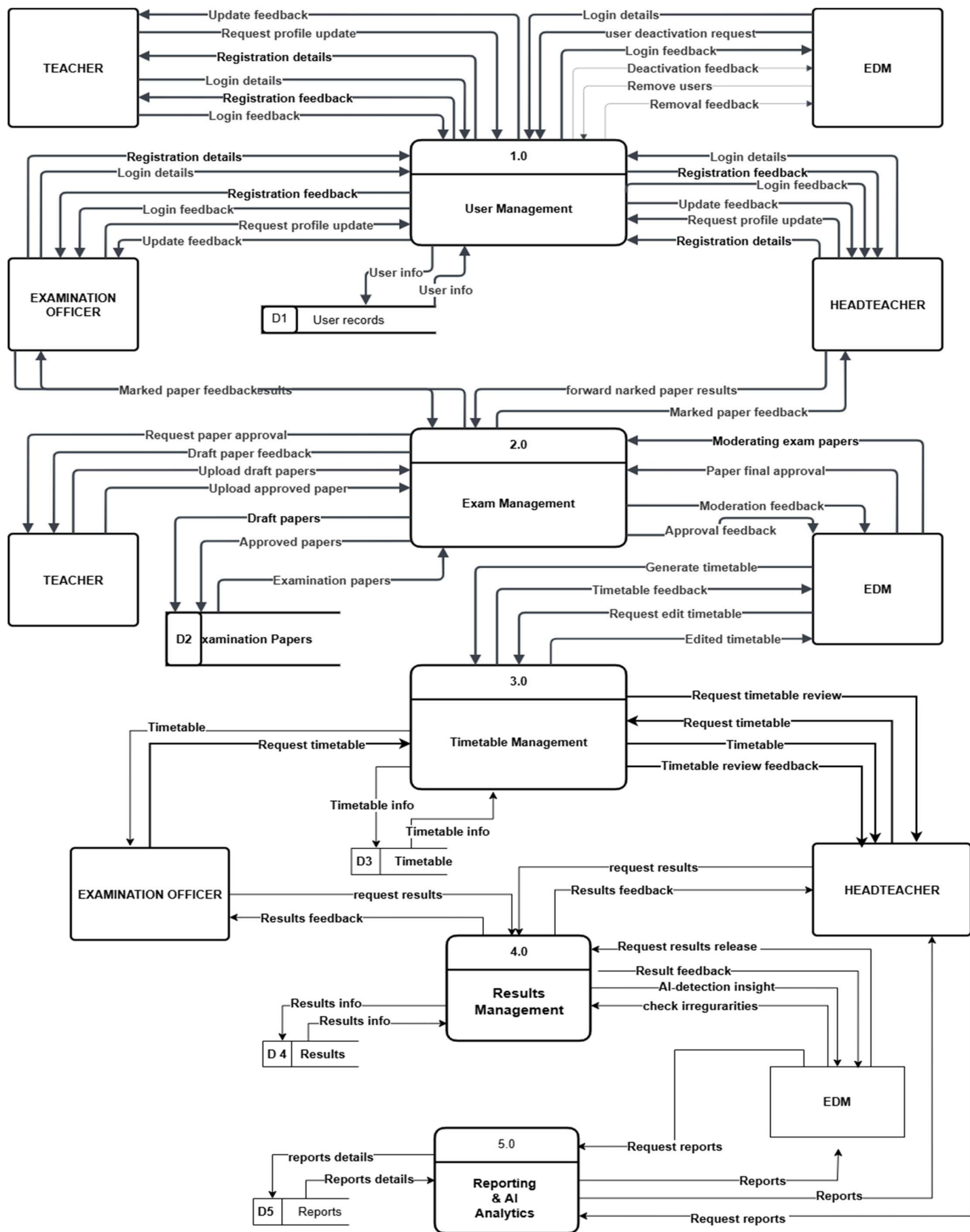
5.1.2 Level zero diagram for the current system



5.1.3 Figure 3: Context diagram for the proposed system



5.1.2 Figure 4: Level zero diagram for the proposed system



6.0 System Requirements

REQ.NO	REQUIREMENT	DESCRIPTION
1	The system shall allow users to log into the system using unique credentials.	The system shall authenticate users based on their assigned roles and deny access to unauthorized functions.
2	The system shall allow Item Writers to upload draft exam papers along with relevant metadata, and securely store all documents.	Documents shall be stored in encrypted storage with restricted access, including metadata such as subject, class, exam type, and date.
3	The system shall notify Moderators of exam papers awaiting review, and allow them to approve or reject papers.	Moderators shall provide comments, corrections, and revision requests for all submissions.
4	The system shall notify the Education Division Manager (EDM) when papers pass moderation, enabling final review.	The EDM shall perform the final approval before the exam release.
5	The system shall provide real-time, role-specific notifications for key actions.	Notifications shall cover submissions, moderation results, approvals, and exam releases.
6	The system shall restrict access to exams until the official release date.	Exams shall remain locked and inaccessible until the scheduled date.
7	The system shall generate reports for all system activities.	Reports shall include submission logs, moderation logs, approvals, and compliance records.
8	The system shall allow administrators to create and manage user accounts and roles.	Administrators shall be able to add, update, deactivate, and delete system users.
9	The system shall prevent unauthorized download or printing of exam papers.	Access restrictions shall maintain exam confidentiality throughout the workflow.

6.2 Interface Requirements

Req. No.	Requirement	Description	External System
2	The system shall validate user information internally.	The system shall verify teacher, headteacher, examination officer, and EDM registration details using internal NED-SEMS records and display a confirmation message when the information is valid.	None (Internal Module)
7	The system shall send automated notifications to users.	The system shall use PHPMailer to send emails for registration confirmation, moderation feedback, paper approvals or rejections, submission alerts, and password reset requests.	PHPMailer (Email Sending Library)

6.3 Non-Functional Requirements

6.3.1 Table 3: System-Related Non-Functional Requirements

Req. No	Non-Functional Requirement	Description	Affected Req. No
1	Security	The system shall protect user accounts by hashing all passwords using BCrypt and allowing only authenticated users to access exam materials. This ensures the confidentiality of sensitive examination documents.	1, 2, 3, 4, 8, 9

2	Performance	With stable internet, the system shall process uploads, approvals, notifications, and report generation within reasonable time. This ensures a smooth experience for teachers, moderators, EDM, and examination officers.	2, 3, 4, 5, 7
3	Accuracy	The system shall store examination papers, user profiles, moderation feedback, and results accurately and consistently to maintain the integrity of examination records.	2, 3, 4, 5, 7
4	Privacy	Each user shall only access information permitted by their role. This prevents unauthorized access to confidential exam materials and internal system data.	1, 2, 3, 4, 6, 9
5	Modifiability	The system shall be developed in a way that supports future updates such as new reports, AI analytics, or SMS notifications without needing major redesign.	1–9

6.3.2 User-Related Non-Functional Requirements

a) Skill Level

Users with basic computer knowledge shall be able to operate the system without requiring technical expertise.

b) Training

System usage training shall take approximately a week for all users to get familiar with the system as long as they know computer usage. A user manual shall also be created which shall show how to perform tasks in the system

7.0 Engineering Plan

7.1 System Lifecycle Constraints

7.1.1 Development Constraints

Development is constrained by the academic-year timeline, which sets fixed deadlines for project activities and limits how many features can be completed. To manage this, the project will follow an Agile approach with structured sprints to ensure steady progress before submission dates. Limited student development resources also require the system to be built using tools that can be implemented and tested reliably within the available time [3] [6] [7].

7.1.5 Maintenance & Retirement Constraints

Maintenance cannot interrupt exam processes, so updates will be tested on a staging server and applied during non-exam periods. Data integrity must also be preserved during updates and eventual system retirement, which will be ensured through backups, rollback procedures, and export tools[11].

7.2 System Delivery

a) Extent of Deliverables

A full working system and instruction manual booklet shall be delivered.

b) Deliverable Formats

The system shall be delivered as a zip of compressed system source files and the instruction manual as a hardcopy booklet.

c) Installation

i. Developer Access to Installed Environment

- Developers shall be granted temporary access to the hosting server for installation, testing, and configuration.
- Access permissions will be role-restricted to prevent exposure of confidential exam content.

ii. Procedures to Replace Existing System

If an older examination system exists, the installation process shall include:

- Backing up all existing exam data and user records.
- Migrating relevant records into the new system (where applicable).
- Installing NED-SEMS on the live server without disrupting ongoing school operations.
- Performing system tests and verification with NED officers before going live.

iii. Phase-In Procedures to Replace Existing System

- Start with a controlled pilot rollout to a small group of end-users.
- Test interoperability with existing data flows, interfaces, and system components.
- Expand usage gradually while running the old and new systems concurrently.
- Collect user feedback and monitor system performance during the transition.
- Fix defects and optimize the system based on real user interactions.

- Decommission the old system once the new system is fully stable and validated.

8.0 Glossary

Term	Definition
AI(Artificial Intelligence)	A field of computer science that enables machines or systems to simulate human intelligence processes such as learning, reasoning, and problem-solving.
Agile Methodology	A flexible and iterative software development approach that focuses on collaboration, user feedback, and continuous improvement through incremental development cycles.
Examination Management System (EMS)	A digital platform that automates processes involved in exam preparation, moderation, administration, and reporting.
NED-SEMS	The proposed Northern Education Division Smart Examination Management System a web-based intelligent platform designed to enhance efficiency, transparency, and security in examination management.
EDM(Education Division Manager)	The head of the education division responsible for overseeing examination activities and general education management within the Northern Division.
Data Analytics	The process of examining data sets to identify patterns, trends, and insights that support informed decision-making.
Performance Trends	Patterns or changes observed in students' academic performance over time.
Examination Irregularities	Any unauthorized actions or practices that compromise the integrity, fairness, or security of examinations.

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